MIL-C-47150(MI)
7 June 1974

SUPERSEDING
MPD-9621B
7 September 1961

# MILITARY SPECIFICATION
# COMPOUND, COATING, PORCELAIN FOR POTTING ELECTRONIC COMPONENTS

This specification is approved for use by all departments and agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the requirements for one type of porcelain casting compound used for potting electronic components.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on the date of invitation for bids, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

| | |
|---|---|
| UU-T-111 | Tape, Paper, Gummed (Sealing and Securing) |
| PPP-B-585 | Boxes, Wood, Wirebound |
| PPP-B-591 | Boxes, Fiberboard, Wood-Cleated |
| PPP-B-601 | Boxes, Wood Cleated-Plywood |

FSC 5970

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| | |
|---|---|
| PPP-B-621 | Boxes, Wood, Nailed and Lock-Corner |
| PPP-B-636 | Box, Fiberboard |
| PPP-C-96 | Cans, Metal, 28 Gage and Lighter |
| PPP-T-76 | Tape, Pressure-Sensitive, Adhesive, Paper, Water-Resistant (for Carton Sealing) |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-109 | Inspection Terms and Definitions |
| MIL-STD-129 | Marking for Shipment and Storage |

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Society for Testing Materials

| | |
|---|---|
| ASTM C 236 | Thermal Conductance and Transmittance of Built-up Sections by Means of the Guarded Hot Box, Method of Test for |



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(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Uniform Classification Committee

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Freight Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60605.)

American Trucking Association Publication

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., 1616 P Street N.W., Washington, D.C. 20036.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

## 3. REQUIREMENTS

3.1 Preproduction sample. The preproduction sample, if required (see 6.2), shall meet all requirements of this specification.

3.2 Materials. The casting compound covered by this specification shall be apowdered ceramic compound which, after the addition of water, cures to form a rigid porcelain material.

3.2.1 Cure. The potting compound described in this specification shall meet the requirements herein when mixed in the proportion of 43 parts of powder to 10 parts of water and cured according to either of the following cure cycles:



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| Time (hours) | Temperature (degrees) |
|---|---|
| 24 | 23 plus or minus 3 Celsius (C) (73 plus or minus 5 Fahrenheit (F)) |
| 4.5 | 74 plus or minus 3 C 165 plus or minus 5 F |

3.3 Physical properties of cured casting compound. The casting compound, after being mixed and cured in accordance with the manufacturer's instructions, shall meet the physical property requirements of this specification. The compound shall exhibit a mild expansion action upon curing.

3.3.1 Thermal conductivity. The cured casting compound shall have a thermal conductivity of at least 10 British thermal units per hour per square foot per degree F per foot of thickness.

3.4 Low-temperature storage. The cured casting compound shall meet the requirements of this specification after storage for 72 hours at minus 62 plus or minus 3 degrees C (minus 80 plus or minus 5 degrees F).

3.5 Workmanship. The workmanship shall be such as to insure a product which is uniform, in conformance with this specification, and properly packaged and packed. The casting compound shall be free from dirt and foreign material. Containers shall be clean, uniformly filled, well sealed, and legibly marked.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the



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specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The classification of inspection shall be as follows:

a. Preproduction inspection.

b. Acceptance inspection.

4.3 Preproduction inspection.

4.3.1 Preproduction sample. A preproduction sample to be inspected under this document shall be representative of the subsequent production and shall meet all the requirements as specified herein.

4.3.2 Sampling. Sampling for preproduction inspection shall be specified by the procuring activity.

4.3.3 Inspection site. Unless otherwise specified, the preproduction inspection site shall be as outlined in 4.1.

4.3.4 Examinations. The preproduction examinations shall consist of all the acceptance examinations specified herein.

4.3.5 Testing. The preproduction tests shall consist of all the acceptance tests specified herein plus the following test:

4.3.5.1 Low temperature storage. The cured casting compound shall be placed in a temperature chamber at a temperature of minus 62 plus or minus 3 degrees C (minus 80 plus or minus 5 degrees F) for a minimum of 72 hours. Upon completion of the 72-hour exposure, the casting compound shall be allowed to stabilize at normal room temperature and then shall be tested for conformance with all the acceptance tests specified herein.

4.4 Acceptance inspection.

4.4.1 Lot size. Lot size shall not exceed that quantity of casting compound produced in one continuous operation by the same manufacturer and manufactured in accordance with the same specification, specification revision,



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and other requirements. Each lot shall be composed of raw materials from homogeneous lots.

4.4.2 Sampling.

4.4.2.1 For examination. Sampling for examination shall be 100 percent of production except as specified in 4.4.2.1.1.

4.4.2.1.1 Sampling for inspection of containers. A random sample of filled containers shall be selected by the Government inspector in accordance with MIL-STD-105 at inspection Level I and acceptable quality level = 2.5 percent to verify compliance with all requirements regarding fill, closure, marking, and other requirements not requiring tests.

4.4.2.2 For testing. Unless otherwise specified, testing shall be performed on a sample taken at random from each lot. The sample shall be of a size sufficient for the performance of the required tests and shall be marked with the manufacturer's name, product number, lot number, and data of manufacture.

4.4.3 Acceptance examinations. Visual examination of the samples specified in 4.4.2.1 shall be conducted for the purpose of determining compliance with the requirements for construction and workmanship, as well as the requirements for preservation, packaging, packing, and marking covered in Section 5.

4.4.3.1 Inspection of containers. Each sample filled container shall be examined for defects of construction of the container and the closure for evidence of leakage and for unsatisfactory marking; each filled container shall also be weighed to determine the amount of contents. Any container in the sample having one or more defects or under required fill shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, the lot represented by the sample shall be rejected. Rejected lots may be resubmitted for Government inspection provided the contractor has removed or repaired all nonconforming containers.

4.4.4 Testing.



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4.4.4.1 Test methods and procedures.

4.4.4.1.1 Test specimens. Five test specimens shall be prepared in accordance with the respective test method requirements for each test specified in 4.7.4.2.

4.4.4.2 Tests.

4.4.4.2.1 Thermal conductivity. Test specimens of casting compound shall be tested for thermal conductivity in accordance with ASTM C 236. Specimens shall be of the form and dimensions specified in the ASTM test method.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. (See 6.2)

5.1.1 Level A.

5.1.1.1 Unit packaging. The casting compound shall be enclosed in a container conforming to PPP-C-96.

5.1.1.2 Intermediate packaging. When intermediate packaging is required, unit packages shall be placed in close-fitting containers conforming to Class WR of PPP-B-636. Separators, partitions, and filler strips conforming to PPP-F-320, shall be used. The intermediate container shall be closed as specified in 5.1.1.3.

5.1.1.3 Closure of fiberboard boxes. Unless otherwise specified, containers conforming to PPP-B-636 shall be closed with tape conforming to PPP-T-76.

5.1.2 Level B (see 6.3.3). Same as Level A.

5.1.3 Level C. Unless otherwise specified, preservation and packaging shall afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity for immediate use.

5.2 Packing. (See 6.2)

5.2.1 Level A. The casting compound shall be packed in overseas-type exterior containers conforming to one of the following specifications: PPP-B-585, PPP-B-591; or PPP-B-621. The container shall be strapped in accordance with the applicable container specification.



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5.2.2 Level B. The casting compound shall be packed in domestic-type exterior containers conforming to one of the following specifications: PPP-B-585; PPP-B-591; PPP-B-601; or PPP-B-621.

5.2.3 Level C. The casting compound shall be packed for shipment in conformance with Uniform Freight Classification Rules and Container Specifications for Rail Shipments or with National Motor Freight Classification Rules and Container Specifications for Truck Shipments.

5.3 Markings. Interior packages and exterior containers shall be marked in accordance with Standard MIL-STD-129 as implemented by the using service.

6. NOTES

6.1 Intended use. The casting compound covered by this specification is intended for use as a thermally-conductive potting compound for electronic components.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether or not a preproduction sample is required (see 3.1).

c. Applicable sampling plan and level of inspection.

d. Selection of applicable levels of preservation, packaging, and packing (see 5.1 and 5.2).

6.3 Definitions.

6.3.1 Service temperature range. The cured casting compound cover by this specification shall remain dimensionally stable and serviceable throughout the temperature range minus 62 to plus 427 C (minus 80 to plus 800 degrees F).

6.3.2 Inspection terms. The definitions of inspection terms shall be in accordance with MIL-STD-109.



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6.3.3 Levels of protection. Level B preservation and packaging is intended to provide economical but limited protection and should be specified only when it is determined to be appropriate.

6.4 Supersession data. This specification contains the requirements of Missile Purchase Description MPD-9621B, dated 7 September 1961.

Custodian:
ARMY-MI

Preparing Activity:
ARMY - MI
Project No. 5970-A469



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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

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